Seminar Report

AI BASED WASTE MANAGEMENT SYSTEM

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ABSTRACT

Waste generation is increasing rapidly in cities because of population growth, urbanisation, and changing lifestyles. Traditional waste management depends on manual segregation, fixed collection schedules, and overflowing community bins. These methods increase pollution, create health risks, increase fuel and labour cost, and cause a large amount of recyclable material to be lost because dry, wet, and hazardous waste get mixed at the source. Proper segregation and timely collection are therefore essential for clean and sustainable cities.

This project focuses on the study of an AI Based Smart Waste Management System using waste images, smart bin sensors, and modern artificial intelligence techniques. It examines approaches such as computer vision, Convolutional Neural Networks (CNN), Machine Learning, Deep Learning, IoT-based smart bins, and route optimisation for garbage collection vehicles. These techniques help identify the type of waste, such as plastic, paper, metal, glass, and organic waste, monitor the fill level of bins, and support automatic waste segregation. The study also considers real-time dashboards, low-cost hardware such as Arduino and ultrasonic sensors, and the use of blockchain and robotic sorting in the future.

The literature review analyses research papers on AI based waste management to understand the methods used, their major findings, and their limitations. It highlights challenges such as high implementation cost, limited quality data, dependence on image quality and lighting conditions, network problems, and lack of public awareness. Overall, AI based smart waste management has the potential to reduce collection time, fuel consumption, operational cost, overflowing bins, and environmental pollution, and to contribute to cleaner, safer, and more sustainable cities.

CHAPTER 1

INTRODUCTION

Every household, market, hospital, hotel, and industry produces waste every day. As cities grow, the amount of municipal solid waste is also increasing, and many local bodies find it difficult to collect, transport, and treat it properly. Waste that is not segregated at the source ends up in landfills, where wet waste, plastic, paper, metal, and hazardous items are mixed together. This pollutes soil, water, and air, spreads diseases, and makes recycling very difficult.

In most cities, waste bins are emptied on a fixed schedule whether they are full or not. Some bins overflow and spread garbage on the road, while other bins are collected when they are almost empty. Workers who segregate waste by hand are exposed to sharp objects, medical waste, and harmful chemicals. Traditional waste management also needs more time, manual work, fuel, and money. These problems show the need for a smarter and more reliable system.

With the development of Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and the Internet of Things (IoT), smart waste management has become a promising research area. A camera and a trained image-classification model can identify the type of waste, such as plastic, paper, metal, glass, or organic waste. Low-cost sensors such as ultrasonic sensors can measure how full a bin is, and the data can be sent to a dashboard so that collection vehicles go only where they are needed. AI can also help find the shortest route for garbage collection vehicles, saving time and fuel.

Several researchers have developed AI based waste management systems. Some studies focus on classifying waste from images using Convolutional Neural Networks (CNN), others combine IoT sensors with cloud platforms to monitor bins in real time, and some use route optimisation to reduce the distance travelled by collection vehicles. Despite these advancements, challenges such as high implementation cost, limited quality data, dependence on image quality and lighting conditions, network problems, and lack of public awareness still exist.

This literature review examines research papers related to AI based waste management. It discusses the techniques, major findings, and limitations presented in these studies. The review also identifies research gaps that can guide the development of a practical and affordable AI Based Smart Waste Management System. The topic also supports the aims of the Swachh Bharat Mission and the Solid Waste Management Rules, 2016.

CHAPTER 2

LITERATURE REVIEW

A literature review studies existing research on a topic. Here, it studies research on AI based waste management that uses image processing, machine learning, deep learning, IoT sensors, and route optimisation. The following papers discuss their techniques, achievements, and limitations.

- **Review of Research Papers**

1. AI-Based Waste Management System

Authors: Prof. Suraj Mahajan

This research explains how AI improves waste management by identifying waste types, monitoring smart bin levels, and optimizing garbage collection routes. It saves time, fuel, and manual effort. Despite challenges like high costs and technical difficulties, AI can help create cleaner, safer, and more sustainable cities.

2. AI-Powered Smart Waste Segregation System

Authors: Prof. Shubhangi Ingale

This research explains how AI uses computer vision and CNN to classify waste into biodegradable, recyclable, and non-recyclable categories. The system automatically separates waste using sensors and motors, reducing manual work and errors while improving recycling efficiency and worker safety.

3. Integration of IoT and AI for Smart Waste Management System

Authors: VinothKumar T, Mr. M. SathishKumar.

This research explains how IoT and AI improve waste management using smart bins, sensors, and cloud platforms. AI predicts waste generation and optimizes collection routes, preventing bin overflow and saving fuel. The system can reduce collection frequency by up to 40% and vehicle travel distance by over 30%, making waste management more efficient.

4. AI-Powered Smart Waste Management System: Optimizing Waste Collection and Recycling through Intelligent Automation

Authors: Ritika Chaudhary, Rishant Rana, Swati Raj Shalu, Ashish Kumar,Amit Raj

This research explains how AI, Machine Learning, IoT sensors, and computer vision improve waste management. The system monitors dustbins, separates waste, and optimizes collection routes, reducing fuel costs, pollution, and bin overflow. Despite challenges like high costs and network issues, future technologies can make waste management smarter and more sustainable.

5. AI-Based Waste Management System (Machine Learning, Deep Learning and IoT)

Authors: Md. Wahidur Rahman, Rahabul Islam, Arafat Hasan, Nasima Islam Bithi.

This research explains how AI, Machine Learning, Deep Learning, and IoT improve waste management. Smart bins monitor waste levels, identify waste types, and optimize collection routes. The system reduces time, fuel consumption, costs, and pollution. Future technologies can make waste management more efficient and sustainable.

6. Classification of Trash for Recyclability Status (2016)

Authors: Mindy Yang, Gary Thung

This study by Mindy Yang and Gary Thung (Stanford University) introduced the TrashNet dataset, containing about 2,500 waste images in six categories: glass, paper, cardboard, plastic, metal, and trash. It helps train AI models for waste classification. However, limited images and plain backgrounds may reduce accuracy in real-world conditions.

7. MobileNetV2: Inverted Residuals and Linear Bottlenecks (2018)

Authors: Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov.

This study by Sandler et al. introduced MobileNetV2, a lightweight deep learning model for mobile and embedded devices. It requires less memory and computation, making it suit able for low-cost smart bins and waste classification. However, its accuracy depends on training image quality and may require further optimization for small devices.

8. AI-Based Waste Management System :Review Smart Cities

Authors: Bingbing Fang, Jiacheng Yu, Zhonghao Chen, Ahmed I. Osman, Mohamed Farghali, Ikko Ihara, Essam H. Hamza, David W. Rooney, Pow-Seng Yap.

This research explains how AI improves waste management by identifying waste types, monitoring smart bin levels, and optimizing collection routes. The system achieved 94% waste classification accuracy and helps reduce collection time, costs, and pollution. Despite challenges like high costs and limited data, AI supports cleaner, safer, and more sustainable cities.

9. AI-Based Waste Management System

Authors: David B. Olawade, Oluwaseun Fapohunda, Ojima Z. Wada, Sunday O. Usman, Abimbola O. Ige, Olawale Ajisafe, Bankole I. Oladapo.

This research explains how AI improves waste management by identifying waste types, monitoring smart bins, and optimizing collection routes. It reduces manual work, saves time and fuel, improves recycling, and reduces pollution. Despite challenges like high costs and limited data, AI helps create cleaner and more sustainable cities.

10. AI-Based Waste Management System

Authors: Aadil Bari, Oroos Arshi, Surajit Mondal

This research explains how AI, IoT, and smart sensors improve waste management. The system achieved 94.3% waste classification accuracy, reduced fuel consumption by 32%, operational costs by 28%, and overflowing bins by 40%. Despite challenges like high costs and maintenance, it supports cleaner, smarter, and more sustainable cities.

11: AI-Based Waste Management System

Authors: Dhanashree Vipul Yevle, Palvinder Singh Mann.

AI-Based Waste Management System uses AI, IoT, and smart sensors to identify waste, monitor bin levels, and optimize collection routes. It saves time and fuel, improves recycling, reduces overflowing bins, and controls pollution. Despite high costs and technical challenges, it helps create cleaner and more sustainable cities.

12. AI-Powered Smart Waste Segregation System

Authors: Prof. Shubhangi Ingale, Vedant Gawali, Sumit Pokharkar, Gopal Payghan

This research presents an AI-powered waste segregation system using Computer Vision, Deep Learning, CNN, and IoT. It classifies waste into biodegradable, recyclable, and non-recyclable categories and automatically directs it into the correct bins. The system reduces manual work, and supports environmental sustainability.

Table No 2.1: Literature review:

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	AI-based waste management using waste-type identification, smart bin level monitoring and collection route optimisation. (Prof. Suraj Mahajan)	Artificial Intelligence; smart bins with level sensors; route-optimisation algorithms.	Not stated in the notes.	Shows how AI identifies waste types, monitors smart bin levels and optimises garbage collection routes. Saves time, fuel and manual effort and supports cleaner, safer and more sustainable cities.	High cost and technical difficulties in implementation.
2	CNN-based automatic waste segregation using computer vision. (Prof. Shubhangi Ingale)	Computer vision; Convolutional Neural Network (CNN); sensors; motors for automatic separation.	Not stated in the notes.	Classifies waste into biodegradable, recyclable and non-recyclable categories and separates it automatically. Reduces manual work and errors, and improves recycling efficiency and worker safety.	Needs sensor and motor hardware. Specific limitations are not stated in the notes.
3	Integration of IoT and AI with smart bins and cloud monitoring. (VinothKumar T, M. SathishKumar)	IoT smart bins; sensors; cloud platforms; AI models for predicting waste generation.	Not stated in the notes.	Predicts waste generation and optimises collection routes to prevent bin overflow and save fuel. Reported up to 40% reduction in collection frequency and over 30% reduction in vehicle travel distance.	Depends on sensors and cloud connectivity. Specific limitations are not stated in the notes.
4	Intelligent automation for waste collection and recycling. (Ritika Chaudhary, Rishant Rana, Swati Raj Shalu, Ashish Kumar, Amit Raj)	Artificial Intelligence; Machine Learning; IoT sensors; computer vision.	Not stated in the notes.	Monitors dustbins, separates waste and optimises collection routes. Reduces fuel cost, pollution and bin overflow. Future technologies can make waste management smarter and more sustainable.	High cost and network problems.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
5	Combined Machine Learning, Deep Learning and IoT waste management. (Md. Wahidur Rahman, Rahabul Islam, Arafat Hasan, Nasima Islam Bithi)	Machine Learning; Deep Learning; IoT smart bins.	Not stated in the notes.	Smart bins monitor waste levels, identify waste types and optimise collection routes. Reduces time, fuel consumption, cost and pollution.	Specific limitations are not stated. Future technologies are needed for better efficiency and sustainability.
6	Benchmark dataset for classifying trash by recyclability (2016). (Mindy Yang, Gary Thung, Stanford University)	TrashNet dataset; image classification models for AI training.	TrashNet: about 2,500 waste images in six categories (glass, paper, cardboard, plastic, metal, trash).	Provides a labelled image dataset so that AI models can be trained to classify waste by recyclability.	Limited number of images and plain backgrounds may reduce accuracy in real-world conditions.
7	Lightweight deep learning model for mobile and embedded devices (2018). (Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov)	MobileNetV2; inverted residuals; linear bottlenecks.	General-purpose model. No waste dataset is stated in the notes.	Provides a model that needs less memory and computation, which makes it suitable for low-cost smart bins and waste classification.	Accuracy depends on training image quality. May need further optimisation for small devices.
8	AI-based waste management review for smart cities. (Bingbing Fang, Jiacheng Yu, Zhonghao Chen, Ahmed I. Osman et al.)	AI; waste-type identification; smart bin level monitoring; route optimisation.	Review paper. No single dataset is stated in the notes.	Explains how AI identifies waste, monitors bins and optimises routes. Reported 94% waste classification accuracy, with lower collection time, cost and pollution.	High cost and limited data.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
9	AI-based waste identification, smart bin monitoring and route planning. (David B. Olawade, Oluwaseun Fapohunda, Ojima Z. Wada et al.)	Artificial Intelligence; smart bins; route optimisation.	Not stated in the notes.	Reduces manual work, saves time and fuel, improves recycling and reduces pollution, helping create cleaner and more sustainable cities.	High cost and limited data.
10	AI, IoT and smart sensor based waste management. (Aadil Bari, Oroos Arshi, Surajit Mondal)	Artificial Intelligence; IoT; smart sensors.	Not stated in the notes.	Achieved 94.3% waste classification accuracy. Reduced fuel consumption by 32%, operational cost by 28% and overflowing bins by 40%.	High cost and maintenance.
11	AI, IoT and smart sensors for waste identification and route optimisation. (Dhanashree Vipul Yevle, Palvinder Singh Mann)	Artificial Intelligence; IoT; smart sensors.	Not stated in the notes.	Identifies waste, monitors bin levels and optimises collection routes. Saves time and fuel, improves recycling, reduces overflowing bins and controls pollution.	High cost and technical challenges.
12	CNN-based smart waste segregation with automatic bin routing. (Prof. Shubhangi Ingale, Vedant Gawali, Sumit Pokharkar, Gopal Payghan)	Computer vision; Deep Learning; CNN; IoT.	Not stated in the notes.	Classifies waste into biodegradable, recyclable and non-recyclable categories and directs it automatically into the correct bins. Reduces manual work and supports environmental sustainability.	Specific limitations are not stated in the notes.

2.2 Summary of Literature Review

The reviewed research papers show that artificial intelligence, IoT sensors, and computer vision can improve waste management. AI models such as CNN can identify the type of waste from images, while ultrasonic sensors and cloud platforms can monitor the fill level of bins in real time. AI-based route planning can reduce collection trips, travelled distance, fuel consumption, and operational cost. Lightweight networks such as MobileNetV2 and datasets such as TrashNet make it possible to build low-cost systems.

Despite these advancements, some challenges remain. High implementation cost, limited and low-quality data, dependence on image quality and lighting conditions, network problems, and lack of public awareness can affect the performance of these systems. Therefore, waste management systems should be tested on diverse real-world waste images and in real bins to improve their reliability and practical use.

2.3 Research Gap

Based on the reviewed studies, the following research gaps have been identified:

Limited real-world testing: Many models are trained on clean images and may perform less effectively on dirty, damaged, or overlapping waste.

Dataset limitations: Some waste categories have fewer training images, which may reduce classification reliability.

Separate solutions: Some systems only classify waste, while others only monitor bin levels; very few combine both in a simple system.

Cost and hardware: Several systems need costly hardware, strong network coverage, or large computing resources.

Public awareness: Even a good system gives poor results if citizens do not use the correct bin or do not understand the system.

The proposed AI Based Smart Waste Management System project aims to study these challenges and propose a low-cost approach that combines image-based waste classification with bin fill-level monitoring through a simple web dashboard.

CHAPTER 3

INFORMATION/CHAPTERS

1. Meaning of AI Based Smart Waste Management

AI Based Smart Waste Management refers to the use of artificial intelligence, sensors, and the Internet of Things to collect, segregate, and dispose of waste in an efficient way. Traditional waste management depends on people to separate waste and on fixed collection schedules. This process is time-consuming, unhygienic, and costly.

With advancements in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and image processing, waste can be identified automatically from digital images. The system analyses an image of the waste item and identifies patterns such as shape, colour, and texture that are associated with different types of waste.

A smart waste management system generally accepts an image of a waste item as input, processes the image, extracts useful features, and classifies it using a trained machine learning or deep learning model. A sensor unit in the bin measures how full the bin is. The final result tells the user the waste category and the correct bin, and it tells the collection staff which bin needs to be emptied.

This technology can support correct segregation, reduce overflowing bins, and help municipal bodies plan collection routes. However, the accuracy of the system depends on image quality, dataset diversity, sensor reliability, and the performance of the trained model. The system should also support the Solid Waste Management Rules, 2016, which require segregation of waste at the source into wet, dry, and hazardous categories.

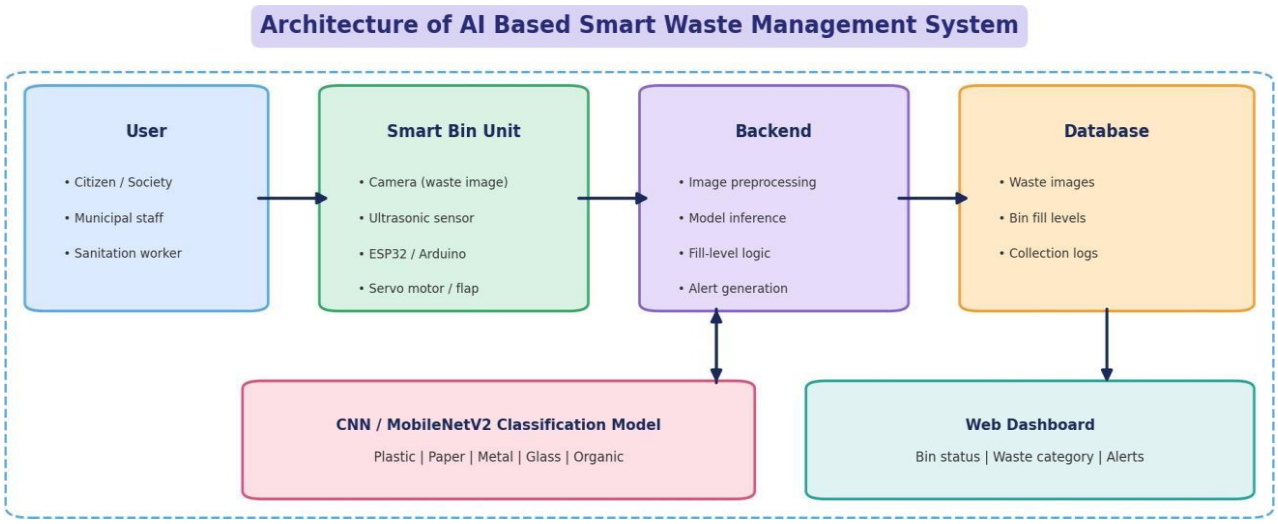


Fig no: 3.1 Architecture of AI Based Smart Waste Management System

Table 3.1: Types of Waste and Suitable Bins

Type of Waste	Examples	Suitable Bin
Wet (biodegradable) waste	Food scraps, vegetable and fruit peels, garden waste	Wet / green bin
Dry waste	Paper, cardboard, plastic, glass, metal, textiles	Dry / blue bin
Hazardous waste	Batteries, chemicals, paint, medical waste	Special hazardous bin
E-waste	Old mobile phones, chargers, circuit boards	E-waste collection point

3.2 Common AI Techniques Used in Smart Waste Management

1. **Machine Learning (ML):** Machine learning algorithms learn patterns from waste images and sensor data and use them to classify waste and predict how fast a bin will fill.
2. **Deep Learning (DL):** Deep learning models automatically learn complex visual patterns from large collections of waste images.
3. **Convolutional Neural Network (CNN):** CNN is widely used for image classification. It can identify features such as shape, colour, and texture of plastic, paper, metal, glass, and organic waste.
4. **Computer Vision and Image Processing:** Image processing techniques improve image quality through resizing, noise removal, and normalization before classification.
5. **Transfer Learning:** Transfer learning uses a pretrained model such as MobileNetV2 and adapts it to a waste classification task. It reduces the amount of training data and time required.
6. **Data Augmentation:** Data augmentation creates modified training images through rotation, flipping, and zooming to improve model generalization.
7. **Route Optimisation:** AI algorithms find the shortest route for collection vehicles using the fill level of bins, which saves time and fuel.

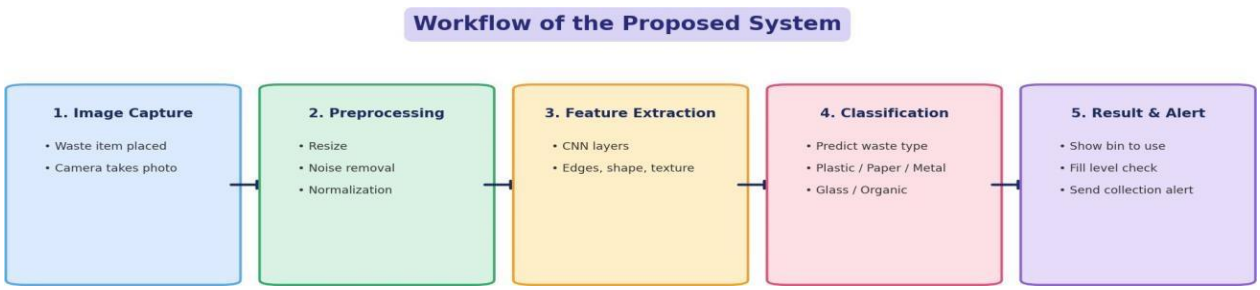


Fig no: 3.2 Workflow of the Proposed System

3.3 Applications of AI Based Smart Waste Management

1. **Automatic Waste Identification:** Identifies the type of waste, such as plastic, paper, metal, glass, or organic waste, from an image.
2. **Smart Bin Monitoring:** Measures the fill level of bins and sends an alert when a bin is almost full.
3. **Waste Segregation:** Directs waste into the correct compartment such as wet, dry, recyclable, or hazardous.
4. **Route Optimisation:** Plans the shortest route for garbage collection vehicles and reduces unnecessary trips.
5. **Recycling Support:** Provides better segregated material to recycling industries and improves recycling rates.
6. **Smart Cities:** Supports clean and sustainable cities through data-based decisions by municipal bodies.
7. **Worker Safety:** Reduces direct human contact with sharp, medical, and hazardous waste.
8. **Public Awareness:** Helps citizens, colleges, and housing societies learn the correct way to segregate waste.

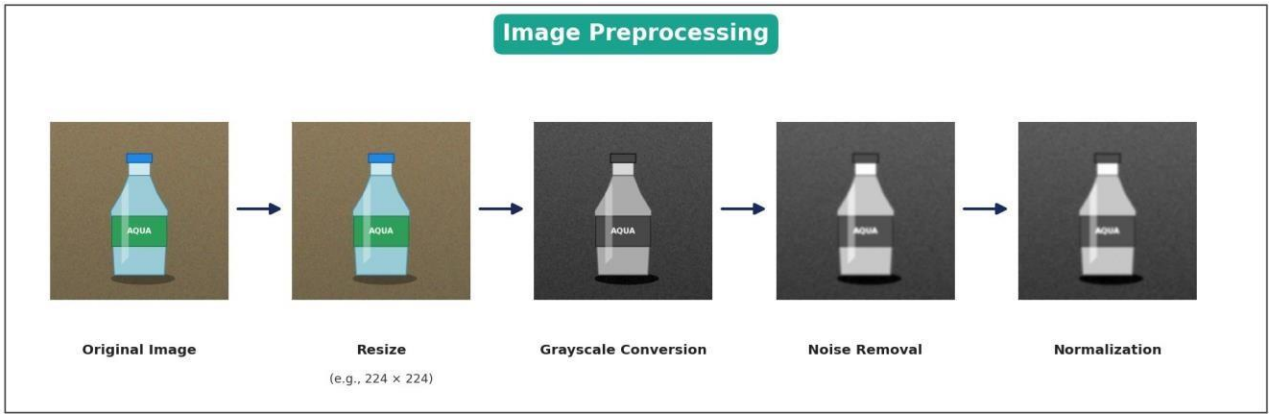


Fig no: 3.3 Image Preprocessing

3.4 Typical Workflow of AI Based Smart Waste Management

Step 1: Image Collection: Images of different types of waste are collected from suitable datasets such as TrashNet or captured using a camera.

Step 2: Image Capture: The user places the waste item in front of the camera of the smart bin, or uploads its image through the system interface.

Step 3: Image Preprocessing: The image is resized to the required dimensions, and suitable preprocessing techniques are applied to prepare it for analysis.

Step 4: Feature Extraction: The system identifies useful visual patterns such as shape, texture, and colour. In a CNN, these features are learned automatically during training.

Step 5: Waste Classification: The trained machine learning or deep learning model analyses the image and predicts the most likely waste category.

Step 6: Result and Bin Status: The system shows the waste category and the correct bin. The ultrasonic sensor measures the fill level, and an alert is generated when the level crosses a set limit, for example 80 percent.

Step 7: Evaluation and Verification: The performance of the model is evaluated using test images and metrics such as accuracy, precision, recall. Doubtful predictions should be checked manually.

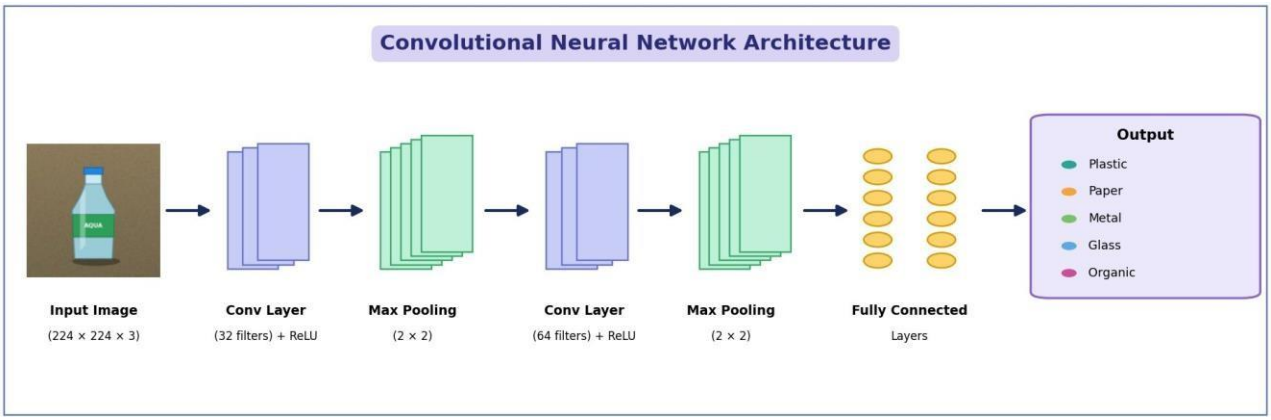


Fig no: 3.4 Convolutional Neural Network Architecture

3.5 Evaluation Metrics

The evaluation metrics are used to measure the performance and effectiveness of the waste classification system. These metrics help determine how accurately the system identifies different types of waste. A combination of different metrics is necessary to understand the overall performance of the model.

1. **Accuracy:** Accuracy measures the percentage of waste images that are correctly classified by the model.
2. **Precision:** Precision measures how many items predicted in a category actually belong to that category. It shows the reliability of predictions.
3. **Recall:** Recall measures how many items of a category are correctly found by the system. A high recall reduces missed items.
4. **F1-Score:** The F1-score represents the balance between precision and recall. It is useful when categories have different numbers of images.
5. **Confusion Matrix:** A confusion matrix shows which categories, for example plastic and glass, are confused with each other.
6. **False-Positive Rate:** This measures how often an item is placed in a wrong category, which can spoil an entire bin of recyclable material.
7. **Processing Time:** Processing time measures how long the system takes to analyse an image and give a result. Lower time improves user experience.
8. **Model Generalization:** This evaluates how well the trained model performs on new waste images that were not included in the training dataset.

These metrics help researchers and developers assess the performance of the system and identify areas requiring improvement. No single metric is sufficient to evaluate the complete system. Therefore, accuracy, precision, recall, F1-score, false-positive rate, and processing time should be considered together. The actual performance.

3.6 Final Waste Classification Result

The final result is the output generated by the system after analysing the waste image and reading the bin sensor. The trained deep learning model processes the image and predicts the waste category, such as plastic, paper, metal, glass, or organic waste.

The result may display the predicted waste type, a confidence score, the suitable compartment or bin, the fill level of the bin, and an alert status. This information helps users dispose of waste correctly and helps collection staff plan their work.

The result is presented through a simple and user-friendly web dashboard so that citizens and municipal staff can easily understand it. If images or location data of users are stored, they should be handled as per the Digital Personal Data Protection (DPDP) Act, 2023, and the Information Technology Act, 2000. Doubtful predictions should be verified manually when necessary.



Fig no: 3.5 Final Waste Classification Result

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Identifies the type of waste automatically using AI.
2. Reduces mixing of wet, dry, and hazardous waste at the source.
3. Prevents overflowing bins by monitoring the fill level.
4. Saves fuel, time, and labour by collecting only full bins.
5. Improves recycling and reduces the load on landfills.
6. Reduces direct human contact with unsafe waste.
7. Uses open-source tools and datasets, which reduces development cost.
8. Can be extended to a mobile app or a city-level dashboard.

4.2 Disadvantages:

1. Accuracy depends on the quality and variety of training images.
2. Dirty, damaged, or overlapping items may be classified wrongly.
3. Sensors and cameras need power, internet, and regular maintenance.
4. Initial hardware and installation cost can be high for large cities.
5. Sensors may give wrong readings because of moisture or sticky waste.
6. Cannot identify waste that is hidden inside bags or covered by other items.
7. Stored images and location data raise privacy and security concerns.
8. The model needs to be retrained when new types of waste or packaging appear.

CHAPTER 5

CONCLUSION

AI based smart waste management has become an important approach for handling the growing amount of waste in cities. The reviewed research papers demonstrate that different techniques, such as computer vision, Convolutional Neural Network (CNN), Machine Learning, Deep Learning, and IoT sensors, can be used to identify the type of waste, monitor bins, and plan collection routes. Among these techniques, deep learning models have shown promising results in automatic waste classification, and one reviewed paper reports 94% accuracy in waste classification.

The studies also highlight the importance of smart bins with ultrasonic sensors and cloud dashboards, which prevent bin overflow and reduce unnecessary collection trips. One reviewed paper reports a possible reduction of up to 40% in collection frequency and over 30% in the distance travelled by collection vehicles. These developments show that AI can reduce collection time, fuel consumption, operational cost, and environmental pollution, and can protect sanitation workers from hazardous waste. In conclusion, AI offers a promising solution for smart waste management. An accurate and reliable system can help segregate waste correctly, reduce overflowing bins, improve recycling, and keep cities clean. In the future, low-cost AI devices, blockchain technology, robotic waste sorting, a mobile application for citizens, and regular model updates with new images can make waste management more efficient, transparent, and sustainable, and can help build cleaner cities, protect the environment, and improve public health.

CHAPTER 6

PROJECT INITIATION

6.1 Description of Problem Statement

Cities produce a large amount of waste every day, and proper waste management is important for public health and a clean environment. However, waste in many cities is mixed at the source, and community bins are emptied on a fixed schedule. This causes overflowing bins, pollution, health risks, high collection cost, and poor recycling. Therefore, there is a need for an automated system that can help identify the type of waste and monitor the fill level of bins quickly and efficiently. Artificial Intelligence, Machine Learning, Deep Learning, image processing, and IoT sensors can be used to analyse waste images and bin levels.

The proposed AI Based Smart Waste Management System allows users to show a waste item to a camera. The system processes the image and uses a trained classification model to predict the waste category. A sensor measures the fill level of the bin, and the result is displayed on a web dashboard with an alert when the bin is nearly full. This project aims to study how AI and sensors can support correct segregation and timely collection of waste and help municipal bodies, colleges, and societies keep their surroundings clean.

6.2 Scope the project:

The purpose of this project is to develop an AI-based smart waste management system that classifies waste using images and monitors dustbin levels using sensors. It uses image processing and deep learning to identify waste categories and displays waste information and bin status on a web dashboard. The system can provide confidence scores and alerts when bins are full. Its performance is evaluated using accuracy, precision, recall, and F1-score. The project focuses on educational and research purposes to support efficient waste segregation and monitoring.

Objectives

- To understand the concept of smart waste management using Artificial Intelligence.
- To study the types of waste and the problems in traditional waste management.
- To study image-based waste classification using machine learning and deep learning.
- To understand the working of Convolutional Neural Networks (CNNs) for image classification.
- To study smart bin monitoring using IoT sensors.
- To study AI-based waste segregation methods.

6.3 Stakeholders

1. **Project Guide:** Provides guidance, monitors progress, and reviews the project work.
2. **Students / Researchers:** Collect information, study research papers, and prepare the report on smart waste management.
3. **Citizens and Households:** Learn the correct bin for each type of waste.
4. **Municipal Corporations and Gram Panchayats:** Plan collection and reduce cost.
5. **Sanitation Workers:** Get safer and more organised work.
6. **Recycling Industries:** Receive better segregated material.
7. **AI / ML Developers:** Develop and evaluate machine learning or deep learning models for waste classification.
8. **Dataset Providers:** Provide waste images and labels required for model training and testing.
9. **Educational Institutions and Housing Societies:** Keep campuses and societies clean and support research and learning.

6.4 Platform / Equipment / Resources Identification

1. Platform

The AI-Based Waste Management System can be developed on Windows or Linux operating systems using Python programming language. Tools such as Visual Studio Code can be used for coding and testing. The system can use TensorFlow/Keras and OpenCV for waste image classification, Flask for backend development, and HTML, CSS, and JavaScript for the web dashboard. The TrashNet dataset can be used to train and evaluate the AI model. IoT sensors can be integrated to monitor dustbin levels and generate alerts when bins are full.

2. Equipment:

- **Computer or Laptop:** For developing and running the application.
- **Processor:** A standard modern processor suitable for Python development.
- **RAM:** Minimum 8 GB recommended; additional memory may be required for running local language models.
- **Internet Connection:** Required for accessing external AI services,downloading libraries.

- **Storage:** Sufficient space for source code, required libraries, and model files, if used.
- **Microsoft Word / Google Docs:** To prepare and format the project report.
- **Microsoft PowerPoint / Google Slides:** To create the seminar presentation.

3. Resources

Resources	Name	Purpose
Operating System	Windows 11	Host and run development tools,.
Programming Language	Python 3.11	To develop the application.
Code Editor	Visual Studio Code	Code editing, debugging, and extension support for development.
AI Waste Detection Tool	YOLO	Detect and classify personally identifiable different types of waste.
Image Processing	OpenCV	To process waste images.
AI Library	TensorFlow	To train and test the AI model
Database	MySQL	To store waste-related information..
Web Framework	Flask	To develop the web application.

BIBLIOGRAPHY

Books

- Tchobanoglous, G., Theisen, H., & Vigil, S. (1993). Integrated Solid Waste Management: Engineering Principles and Management Issues. McGraw-Hill.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- Solid Waste Management — Relevant Educational Reference Books.
- Environmental Science — Relevant Educational Reference Books.
- Integrated Solid Waste Management — George Tchobanoglous and Hilary Theisen.
- Artificial Intelligence: A Modern Approach — Stuart Russell and Peter Norvig

Websites

- Google Scholar – https://scholar.google.com/
- IEEE Xplore Digital Library – https://ieeexplore.ieee.org/
- ScienceDirect – https://www.sciencedirect.com/
- Frontiers in Plant Science – https://www.frontiersin.org/journals/plant-science
- PlantVillage Dataset – https://plantvillage.psu.edu/
- TensorFlow Documentation – https://www.tensorflow.org/
- PyTorch Documentation – https://pytorch.org/docs/stable/
- OpenCV Documentation – https://docs.opencv.org/
- Scikit-learn Documentation – https://scikit-learn.org/stable/

REFERENCES

1. Smart Waste Management: A Paradigm Shift Enabled by Artificial Intelligence

Authors: David B. Olawade, Oluwaseun Fapohunda, Ojima Z. Wada, Sunday O. Usman, Abimbola O. Ige, Olawale Ajisafe, and Bankole I. Oladapo. Link:S2949750724000385.htm

2. AI-Powered Smart Waste Segregation System

Authors: Prof. Shubhangi Ingale, Vedant Gawali, Sumit Pokharkar, and Gopal Payghan. Journal: International Journal of Engineering Research & Technology (IJERT), Volume 15, Issue 06, June 2026.

Link: https://www.irjet.net/archives/V12/i11/IRJET-V12I1153.pdf

3. Integration of IoT and AI for Smart Waste Management System

Authors: VinothKumar T and M. SathishKumar. Journal: International Research Journal of Engineering and Technology (IRJET), Volume 12, Issue 11, November 2025.

4. AI-Powered Smart Waste Management System:

Authors: Ritika Chaudhary, Rishant Rana, Swati Raj Shalu, Ashish Kumar, Amit Raj, and Anmol Bhardwaj.

Publication: Proceedings of the International Conference on Innovative Computing & Communication (ICICC 2024), 2025. Optimizing Waste Collection and Recycling through Intelligent Automation

Link: https://doi.org/10.2139/ssrn.5241203

5. Intelligent Waste Management System Using Deep Learning with IoT

Authors: Md. Wahidur Rahman, Rahabul Islam, Arafat Hasan, Md. Mahmodul Hasan, Nasima Islam Bithi, and Mohammad Motiur Rahman. Journal: Journal of King Saud University – Computer and Information Sciences, Volume 34, Issue 5, Pages 2072–2087, 2022. Link: S1319157820304547.htm

6. M. Yang and G. Thung, "Classification of Trash for Recyclability Status," CS229 Project Report, Stanford University

7. M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," CVPR, 2018. https://arxiv.org/abs/1801.04381

8. B. Fang et al., "Artificial intelligence for waste management in smart cities: a review," Environmental Chemistry Letters, 2023. https://doi.org/10.1007/s10311-023-01604-3

9.D. B. Olawade, O. Fapohunda, O. Z. Wada, S. O. Usman, A. O. Ige, O. Ajisafe, B. I. Oladapo, "Smart waste management: A paradigm shift enabled by artificial intelligence," Waste Management Bulletin, vol. 2, no. 2, pp. 244–263, 2024. https://doi.org/10.1016/j.wmb.2024.05.001 (doaj)

10.A. Bari, O. Arshi, S. Mondal, "Advancements in waste management: a comprehensive review of Artificial Intelligence applications in smart cities." https://journal.hep.com.cn/scasc/EN/10.1007/s44268-026-00088-8 (hep)

11. S. Ingale et al., "AI-Powered Smart Waste Segregation System," Dept. of Computer Engineering, Ajeenkya D. Y. Patil. (IJERT) Link: https://www.ijert.org/?p=55307

SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960212	388520	Vaishnavi Ravindra Kathale
2	24610960231	388539	Divya Ashok Patil
3	24610960238	388546	Kalpesh Sanjay Patil
4	24610960244	388551	Komal Chandrashekhar Patil

Formative Assessment

Sr. No.	Criteria	Weightage	Obtained
1	Selection Topic / Theme of seminar	5	
2	Literature review and data presentation	5	
3	Quality of Preparation and innovativeness	5	
4	Q-A handling	5	
5	Time Management	5	
6	Seminar Presentation report	10	
7	Selection of Theme of Problem Statement and its innovativeness	5	
8	Stages of development of Action plan	5	
9	Prototyping	5	
	Total	**50**	
	Scaled to 25		

Sign :

Name : **Shraddha S Hajari**

(Course Expert/s)

References

- Google Scholar – https://scholar.google.com/
- IEEE Xplore Digital Library – https://ieeexplore.ieee.org/
- ScienceDirect – https://www.sciencedirect.com/
- TrashNet Dataset – https://github.com/garythung/trashnet
- Kaggle – https://www.kaggle.com/
- TensorFlow Documentation – https://www.tensorflow.org/
- PyTorch Documentation – https://pytorch.org/docs/stable/
- OpenCV Documentation – https://docs.opencv.org/
- Scikit-learn Documentation – https://scikit-learn.org/stable/